

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Classical Mechanics

L-T-P: 3-1-0

Credits: 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Principle of virtual work, D'Alembert's Principle, Phase space, Generalized coordinates, Lagrangian, Constraints, Principle of least action, Lagrangian equations of motion, Lagrange multipliers and applications. (16)

Legendre transformation, Hamilton's canonical equations of motion, Routh's procedure, canonical transformations, Poisson brackets and equations of motion. Canonical invariants; infinitesimal contact transformations and conservation laws, angular momentum PB relations, Liouville's theorem. (16)

Hamilton-Jacobi theory, characteristic function, action-angle variables; connections with geometrical optics and wave mechanics. (10)

Reference Books:

1. Classical Mechanics: H. Goldstein

2. Mechanics: L.D. Landau and E.M. Lifshitz

3. Classical Mechanics: N. Rana and P. Joag.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Principle of virtual work, D'Alembert's Principle, Phase space, Generalized coordinates, Lagrangian, Constraints, Principle of least action, Lagrangian equations of motion, Lagrange multipliers and applications.	16
2.	Legendre transformation, Hamilton's canonical equations of motion, Routh's procedure, canonical transformations, Poisson brackets and equations of motion. Canonical invariants; infinitesimal contact transformations and conservation laws, angular momentum PB relations, Liouville's theorem.	16
3.	Hamilton-Jacobi theory, characteristic function, action-angle variables; connections with geometrical optics and wave mechanics.	10
Total number of hours		42